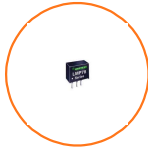


You may
also be
interested in



LMP78_1.2-1.0 by 

Non-Isolated PoL Module DC DC Converter 1 Output 1.2 V
1A 4.5V-42 V Input

[View this part](#)

LM2841YMK-ADJL/NOPB | Texas Instruments Datasheet



SNVS540K – MARCH 2009 – REVISED APRIL 2019

LM284x SIMPLE SWITCHER® 4.5-V to 42-V Input, 0.1-, 0.3-, or 0.6-A Output Step-Down DC/DC Regulator in Thin SOT

1 Features

- Input voltage 4.5 V to 42 V
- Output current options of 100 mA, 300 mA, and 600 mA
- Feedback pin voltage of 0.765 V
- 550-kHz (X) or 1.25-MHz (Y) switching frequency
- Low shutdown I_Q : 16- μ A typical
- Short-circuit protected
- Internally compensated
- Soft-start circuitry
- Small overall solution size (SOT-6L package)
- Create a custom design using the LM2840 (or LM2841/42) with the

2 Applications

- Battery-powered equipment
- Industrial distributed power applications
- Portable media players
- Portable hand-held instruments

3 Description

The LM284x SIMPLE SWITCHER™ devices are PWM DC/DC buck (step-down) regulators. With an input range from 4.5 V to 42 V, they are suitable for a wide range of applications, such as power conditioning from unregulated sources. They feature a low $R_{DS(on)}$ (0.9- Ω typical) internal switch for maximum efficiency (85% typical). Operating frequency is fixed at 550 kHz (X option) and 1.25 MHz (Y option), allowing the use of small external components while still being able to have low output voltage ripple. Soft start can be implemented using the shutdown (SHDN) pin with an external RC circuit allowing the user to tailor the soft-start time to a specific application.

The LM2840 is optimized for up to 100 mA, the LM2841 for up to 300 mA, and the LM2842 for up to 600-mA load currents. They all have a 0.765-V nominal feedback voltage.

Additional features include: thermal shutdown, V_{IN} undervoltage lockout, and gate-drive undervoltage lockout. The LM284x are available in a low-profile SOT-6L package.

Device Information⁽¹⁾

PART NUMBER	PACKAGE	BODY SIZE (NOM)
LM2840, LM2841, LM2842	SOT (6)	1.60 mm × 2.90 mm

For more information on these packages, see the orderable addendum at